

[Submission date]

Editor-in-Chief

Technological Forecasting and Social Change

Dear Editor,

We submit the full-length research article “Re-auditing legacy standards after technological change: a pre-specified protocol for static benchmarks and transition-evidence gates” for consideration in *Technological Forecasting and Social Change*.

Changed sensing, prediction, computation, communication, and control capabilities create a reassessment problem: they alter feasible technical comparators without by themselves establishing that an incumbent should be replaced. The manuscript addresses that problem with a pre-specified Legacy Standard Audit separating a historically defensible incumbent problem, a current frictionless comparator, and a transition-evidence gate.

The empirical demonstration began from 122 candidates in 12 domains and froze 7 heterogeneous cases before confirmatory estimation. The validation is intentionally heterogeneous: it retains material oracle labels, small or uncertain results, an analytic null, and insufficient-evidence cases. Results are not pooled because objectives and units are incomparable, and the manuscript makes no prevalence or general causal claim.

The protocol's governance contribution is an explicit barrier against treating technical improvement as redesign evidence. Simulated, perfect-information, and proxy comparisons are not treated as deployment effects, and no redesign recommendation is made because no case has Level A transition evidence. The work fits the journal because specific technological developments—sensing, computation, communication, prognostics, interval metering, and adaptive control—are the central mechanisms that alter feasible decision sets. The paper connects these mechanisms to standards governance, transition costs, and policy implications.

The manuscript is original, is not under consideration elsewhere, and uses public secondary data and simulations. All authors must confirm authorship, conflicts, and the final declarations before submission.

Sincerely,

[Corresponding author name and affiliation]